

# ROSKILDE AP, Denmark

WMO: 061700

Lat: 55.587N

Lon: 12.136E

Elev: 45

StdP: 100.79

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.1       | -7.0 | -11.4                         | 1.4 | -7.3 | -9.4 | 1.7 | -5.6 | 15.0                  | 5.4  | 13.6 | 4.9  | 2.7                   | 350  | 0.663 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.3                    | 26.0            | 18.4 | 24.2 | 18.0 | 22.8 | 17.4 | 20.0                | 23.6 | 19.0 | 22.6 | 18.1 | 21.5 | 4.6                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.8                            | 13.7 | 21.7 | 17.8 | 12.8 | 20.8 | 16.8 | 12.0 | 19.8 | 57.4          | 23.8 | 54.0 | 22.7 | 51.1 | 21.4 | 23.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 12.2              | 10.8 | 9.7 | -13.3                      | 29.0 | 3.7                | 1.9 | -16.0                                              | 30.4 | -18.1      | 31.5 | -20.2      | 32.5 | -22.8      | 33.9 |  |  |
|                   |      |     | -13.5                      | 21.6 | 3.6                | 1.0 | -16.2                                              | 22.4 | -18.3      | 23.0 | -20.4      | 23.5 | -23.0      | 24.2 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 8.4    | 0.6   | 0.9   | 2.9   | 7.1   | 11.1  | 14.3  | 17.0  | 16.9  | 13.5  | 9.2   | 5.0   | 2.0  |      |
|                                                                                  | DBStd                                                                            | 6.71   | 3.87  | 3.74  | 3.47  | 3.28  | 3.09  | 2.63  | 2.64  | 2.55  | 2.53  | 3.03  | 3.21  | 3.98 |      |
|                                                                                  | HDD10.0                                                                          | 1345   | 292   | 256   | 221   | 99    | 23    | 1     | 0     | 0     | 3     | 51    | 151   | 248  |      |
|                                                                                  | HDD18.3                                                                          | 3655   | 550   | 489   | 479   | 338   | 224   | 124   | 59    | 59    | 144   | 283   | 399   | 506  |      |
|                                                                                  | CDD10.0                                                                          | 769    | 0     | 0     | 1     | 11    | 58    | 130   | 217   | 215   | 109   | 27    | 2     | 0    |      |
|                                                                                  | CDD18.3                                                                          | 38     | 0     | 0     | 0     | 0     | 1     | 3     | 18    | 16    | 1     | 0     | 0     | 0    |      |
|                                                                                  | CDH23.3                                                                          | 247    | 0     | 0     | 0     | 1     | 4     | 27    | 128   | 84    | 3     | 0     | 0     | 0    |      |
|                                                                                  | CDH26.7                                                                          | 29     | 0     | 0     | 0     | 0     | 0     | 4     | 14    | 11    | 0     | 0     | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 4.8    | 5.5   | 5.5   | 5.3   | 4.8   | 4.3   | 4.4   | 4.0   | 4.2   | 4.5   | 4.8   | 4.9   | 5.1  |      |
| Precipitation                                                                    | PrecAvg                                                                          | 630    | 48    | 37    | 35    | 33    | 45    | 62    | 60    | 71    | 63    | 65    | 56    | 55   |      |
|                                                                                  | PrecMax                                                                          | 807    | 92    | 86    | 79    | 77    | 88    | 139   | 154   | 189   | 152   | 128   | 121   | 104  |      |
|                                                                                  | PrecMin                                                                          | 423    | 3     | 6     | 8     | 5     | 12    | 1     | 4     | 16    | 14    | 26    | 12    | 25   |      |
|                                                                                  | PrecStd                                                                          | 94     | 23    | 16    | 19    | 16    | 19    | 32    | 36    | 43    | 38    | 27    | 25    | 21   |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 9.4   | 10.1  | 13.9  | 19.9  | 23.7  | 26.0  | 28.2  | 28.1  | 23.2  | 17.5  | 13.1  | 10.9 |      |
|                                                                                  |                                                                                  | MCWB   | 8.5   | 8.4   | 9.3   | 13.6  | 16.9  | 18.5  | 19.5  | 19.3  | 17.2  | 14.7  | 11.9  | 9.9  |      |
|                                                                                  | 2%                                                                               | DB     | 7.9   | 8.6   | 11.1  | 16.8  | 21.1  | 23.5  | 26.2  | 25.5  | 21.0  | 15.8  | 11.8  | 9.6  |      |
|                                                                                  |                                                                                  | MCWB   | 7.3   | 7.5   | 7.8   | 11.7  | 15.0  | 17.4  | 18.4  | 18.6  | 16.4  | 13.6  | 10.7  | 8.7  |      |
|                                                                                  | 5%                                                                               | DB     | 6.9   | 7.2   | 9.2   | 14.7  | 19.1  | 21.5  | 24.8  | 23.8  | 19.1  | 14.5  | 10.6  | 8.2  |      |
|                                                                                  |                                                                                  | MCWB   | 6.2   | 6.2   | 7.0   | 10.3  | 13.8  | 16.3  | 18.1  | 18.1  | 15.5  | 12.8  | 9.6   | 7.5  |      |
|                                                                                  | 10%                                                                              | DB     | 5.8   | 6.1   | 7.9   | 12.8  | 17.2  | 19.8  | 23.0  | 22.1  | 17.9  | 13.4  | 9.3   | 7.1  |      |
|                                                                                  |                                                                                  | MCWB   | 5.1   | 5.2   | 6.3   | 9.1   | 12.8  | 15.1  | 17.6  | 17.3  | 14.7  | 12.0  | 8.6   | 6.4  |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 8.7   | 8.8   | 10.3  | 14.3  | 17.7  | 20.3  | 21.3  | 21.2  | 18.6  | 15.3  | 12.2 | 10.1 |
|                                                                                  |                                                                                  |        | MCDB  | 9.2   | 9.4   | 12.7  | 18.5  | 22.2  | 24.6  | 25.4  | 24.7  | 21.7  | 16.8  | 12.8 | 10.4 |
| 2%                                                                               |                                                                                  | WB     | 7.2   | 7.6   | 8.5   | 12.3  | 16.0  | 18.2  | 20.2  | 19.9  | 17.2  | 14.1  | 10.9  | 8.8  |      |
|                                                                                  |                                                                                  | MCDB   | 7.6   | 8.3   | 10.2  | 15.8  | 19.5  | 22.0  | 24.1  | 23.1  | 19.8  | 15.3  | 11.6  | 9.3  |      |
| 5%                                                                               |                                                                                  | WB     | 6.1   | 6.3   | 7.4   | 10.8  | 14.5  | 17.0  | 19.2  | 18.9  | 16.2  | 13.1  | 9.7   | 7.5  |      |
|                                                                                  |                                                                                  | MCDB   | 6.6   | 7.0   | 8.7   | 13.7  | 18.0  | 20.3  | 22.9  | 22.3  | 18.5  | 14.3  | 10.3  | 8.0  |      |
| 10%                                                                              |                                                                                  | WB     | 5.0   | 5.1   | 6.2   | 9.6   | 13.3  | 15.8  | 18.2  | 18.0  | 15.3  | 12.3  | 8.7   | 6.5  |      |
|                                                                                  |                                                                                  | MCDB   | 5.5   | 5.9   | 7.6   | 12.1  | 16.5  | 18.9  | 21.8  | 21.3  | 17.3  | 13.4  | 9.2   | 6.9  |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 4.3   | 4.6   | 6.4   | 8.3   | 9.1   | 9.0   | 9.3   | 8.7   | 7.4   | 5.9   | 4.5  | 4.3  |
|                                                                                  |                                                                                  |        | MCDBR | 4.3   | 5.4   | 8.6   | 11.0  | 11.9  | 11.2  | 12.1  | 10.7  | 9.0   | 6.5   | 5.0  | 4.5  |
|                                                                                  | MCWBR                                                                            |        | 4.2   | 4.3   | 6.0   | 6.6   | 7.0   | 6.3   | 5.8   | 5.4   | 5.4   | 4.9   | 4.3   | 4.3  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 4.4   | 5.1   | 7.9   | 10.1  | 10.4  | 10.3  | 10.2  | 9.3   | 8.0   | 6.0   | 4.7   | 4.3  |      |
|                                                                                  |                                                                                  | MCWBR  | 4.3   | 4.4   | 6.0   | 6.5   | 6.7   | 6.5   | 5.7   | 5.4   | 5.2   | 4.8   | 4.3   | 4.3  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.314  | 0.335 | 0.359 | 0.393 | 0.388 | 0.382 | 0.404 | 0.404 | 0.374 | 0.355 | 0.335 | 0.303 |      |      |
|                                                                                  | taud                                                                             | 2.290  | 2.312 | 2.304 | 2.266 | 2.340 | 2.383 | 2.341 | 2.349 | 2.415 | 2.405 | 2.331 | 2.262 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 597    | 729   | 802   | 825   | 854   | 863   | 833   | 807   | 783   | 700   | 564   | 511   |      |      |
|                                                                                  | Edh at Noon                                                                      | 55     | 77    | 98    | 117   | 115   | 112   | 114   | 107   | 88    | 70    | 53    | 45    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 0.45   | 1.05  | 2.46  | 4.17  | 5.36  | 5.73  | 5.32  | 4.27  | 3.02  | 1.49  | 0.57  | 0.31  |      |      |
|                                                                                  | RadStd                                                                           | 0.08   | 0.17  | 0.26  | 0.50  | 0.52  | 0.43  | 0.62  | 0.30  | 0.29  | 0.21  | 0.10  | 0.03  |      |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only            |  | +0.42 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -140    | N/A     | N/A     |
| Regional (66 neighbors) |  | +0.40 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -161    | N/A     | N/A     |

Nomenclature: See separate page